winding carries the input current, $I_L = I_a = I_{se}$. The series field winding is relatively small in number to give minimum value of resistance drop.

7.9.2 DC Shunt Motor

In DC shunt motor, the field winding is connected parallel (shunt) with the armature as shown in fig 7.15. The field winding has a large number of turns and relatively smaller cross sectional area. Therefore, the shunt field winding has high resistance and hence shunt field current is relatively small compared with armature current. The speed of DC shunt motor is almost constant during no-load to full-load.

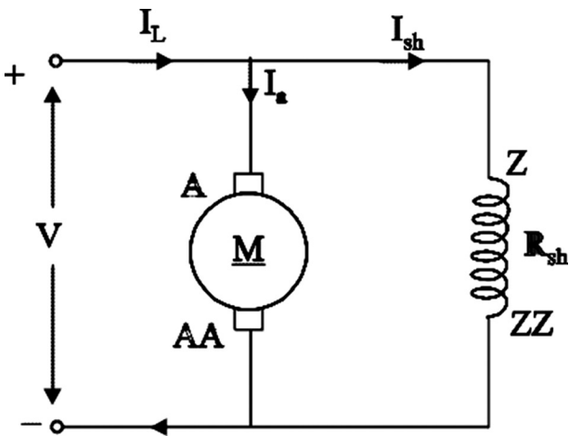


Fig 7.15 *Equivalent Circuit of DC Shunt Motor*

7.9.3 DC Compound Motor

In compound motors both series and shunt windings are connected according to their name of long shunt and short shunt compound motor.

a) Short Shunt Compound Motor

In short shunt compound motor, the series field winding is connected in series with the parallel combination of armature and shunt field winding. The connection diagram for this compound motor is as shown in Fig 7.16.

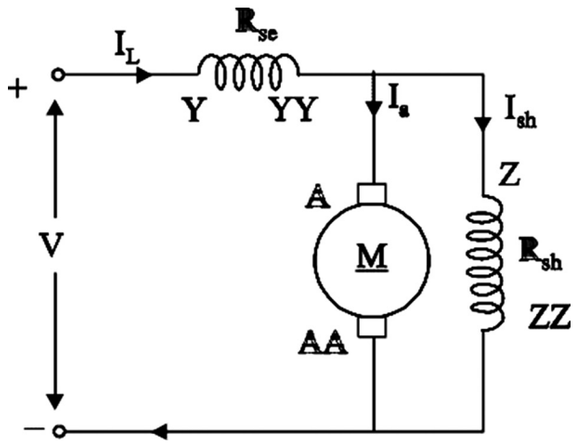


Fig 7.16 *Equivalent Circuit of DC Short Shunt Compound Motor*

b) Long Shunt Compound Motor

In long shunt compound motor, the shunt field winding is connected in parallel with the series combination of series field winding and armature winding.

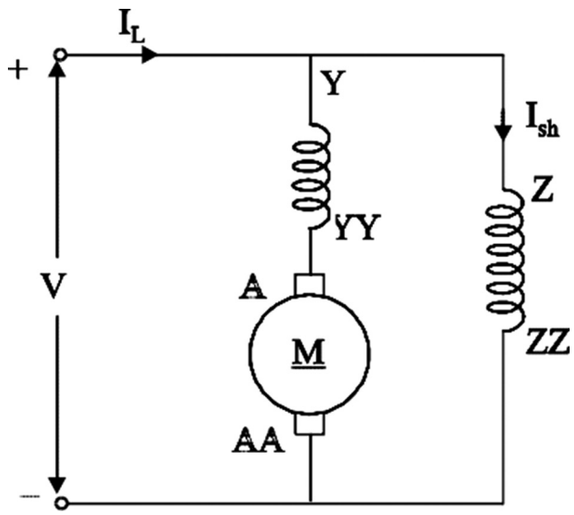


Fig 7.17 *Equivalent Circuit of DC Long Shunt Compound Motor.*

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